Drone Human Detection & Autonomous Tracking System (YOLO + DJI Tello)

Project Overview

This project implements an AI-powered human detection and autonomous tracking system using the DJI Tello drone. The objective is to combine computer vision (YOLO) with real-time drone control, enabling the drone to detect a person, track them smoothly, and capture video/images automatically.
If no human is detected, the drone performs a predefined search pattern to locate one.

This project demonstrates the integration of vision AI with robotics and can be used for surveillance, smart tracking, and human-robot interaction experiments.

Key Features

- Real-time human detection using a custom-trained YOLO model
- Smooth autonomous tracking of a detected human
- Automatic video recording and image capture
- Search mode when no human is detected
- Works with DJI Tello using dji tellopy
- Easy-to-run Python scripts

Technology Stack

- Python 3.x
- YOLOv8 / YOLOv5 (custom human detection model)
- OpenCV
- djitellopy (Tello drone SDK)
- NumPy
- Torch / Ultralytics YOLO

How the System Works

1. Human Detection

A YOLO model trained on human datasets or custom images identifies:

- bounding box
- confidence score
- coordinates

2. Drone Tracking Logic

If a human is detected:

- The drone adjusts yaw, height, and forward/backward movement
- The bounding box center is used for alignment
- A threshold buffer prevents over-movement

3. Search Pattern

If **no human is detected**:

1. Move forward for 20 seconds
2. Rotate/shift right for 20 seconds
3. Repeat until a full cycle is completed

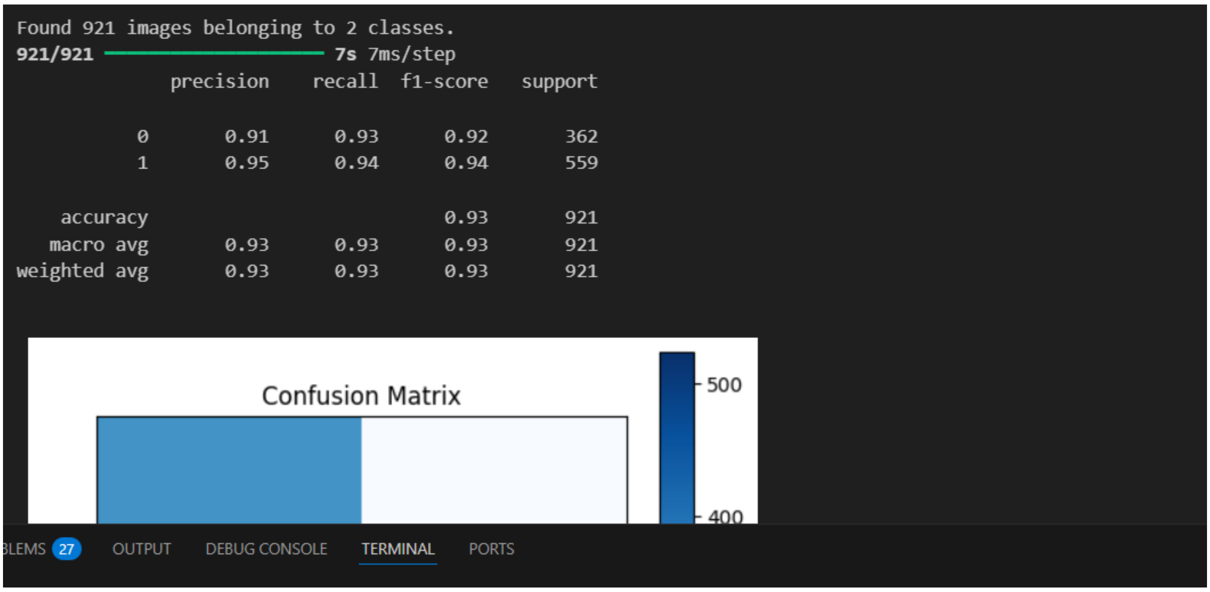
4. Media Capture

```
[1]                                                                                        Python
... Found 738 images belonging to 2 classes.
    Found 183 images belonging to 2 classes.
    c:\Users\bkous\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\convolutional\base_conv.py:107: UserWarning: Do not pass an `input_shape`
      super().__init__(activity_regularizer=activity_regularizer, **kwargs)
    c:\Users\bkous\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\trainers\data_adapters\py_dataset_adapter.py:121: UserWarning: Your `PyDataset`
      self._warn_if_super_not_called()
    Epoch 1/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 19s 390ms/step - accuracy: 0.5510 - loss: 1.2419 - val_accuracy: 0.6066 - val_loss: 0.6682
    Epoch 2/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 189ms/step - accuracy: 0.6225 - loss: 0.6828 - val_accuracy: 0.6066 - val_loss: 0.6523
    Epoch 3/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 183ms/step - accuracy: 0.6553 - loss: 0.6256 - val_accuracy: 0.6230 - val_loss: 0.6032
    Epoch 4/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 184ms/step - accuracy: 0.7640 - loss: 0.5108 - val_accuracy: 0.6612 - val_loss: 0.5373
    Epoch 5/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 177ms/step - accuracy: 0.8366 - loss: 0.4079 - val_accuracy: 0.6612 - val_loss: 0.5783
    Epoch 6/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 188ms/step - accuracy: 0.8594 - loss: 0.3497 - val_accuracy: 0.6721 - val_loss: 0.6745
```

```
ss2.ipynb > # Import necessary libraries
Generate  + Code  + Markdown | Run All  Clear All Outputs | Outline  ...
    Epoch 6/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 188ms/step - accuracy: 0.8594 - loss: 0.3497 - val_accuracy: 0.6721 - val_loss: 0.6745
    Epoch 7/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 182ms/step - accuracy: 0.9083 - loss: 0.2477 - val_accuracy: 0.7213 - val_loss: 0.8587
    Epoch 8/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 179ms/step - accuracy: 0.9252 - loss: 0.1961 - val_accuracy: 0.6667 - val_loss: 0.8026
    Epoch 9/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 177ms/step - accuracy: 0.9486 - loss: 0.1478 - val_accuracy: 0.7049 - val_loss: 0.9994
    Epoch 10/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 180ms/step - accuracy: 0.9783 - loss: 0.0737 - val_accuracy: 0.7104 - val_loss: 1.4903
    Epoch 11/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 178ms/step - accuracy: 0.9788 - loss: 0.0622 - val_accuracy: 0.6448 - val_loss: 1.9440
    Epoch 12/20
    47/47 ━━━━━━━━━━━━━━━━━━━━ 8s 176ms/step - accuracy: 0.9647 - loss: 0.1322 - val_accuracy: 0.6557 - val_loss: 1.5427
    Epoch 13/20
    ...
    47/47 ━━━━━━━━━━━━━━━━━━━━ 9s 182ms/step - accuracy: 0.9960 - loss: 0.0135 - val_accuracy: 0.6667 - val_loss: 2.4041
    12/12 ━━━━━━━━━━━━━━━━━━━━ 2s 130ms/step - accuracy: 0.6593 - loss: 2.3556
    Validation Loss: 2.4041478633880615
    Validation Accuracy: 0.6666666865348816
```

- Drone records live video

- Automatically takes pictures when a person is centered for some time.



```
Found 921 images belonging to 2 classes.
921/921 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step
              precision    recall  f1-score   support

           0       0.91      0.93      0.92       362
           1       0.95      0.94      0.94       559

    accuracy                           0.93       921
   macro avg       0.93      0.93      0.93       921
weighted avg       0.93      0.93      0.93       921
```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How to Run the Project

1.Install Dependencies

pip install -r requirements.txt

2.**Connect to the Tello drone WiFi**

WiFi Name: TELLO-DJI

3.Run the main script

ss2.ipynb